

Complete LocalStack Setup Guide (Windows PowerShell)

Connection Details

Property	Value
Endpoint URL	http://localhost:4566
Region	ap-northeast-1 (Tokyo)
Access Key ID	test
Secret Access Key	test
S3 Bucket Name	my-test-bucket

Prerequisites

- * Install Docker Desktop: <https://www.docker.com/products/docker-desktop/>
- * Install AWS CLI: <https://aws.amazon.com/cli/>
- * Verify AWS CLI: `aws --version`

Step 1 - Start LocalStack

Free Version (localstack:3.0.0):

```
docker run -d --name localstack_test \  
-e SERVICES=s3,ses \  
-e DEFAULT_REGION=ap-northeast-1 \  
-p 4566:4566 \  
localstack/localstack:3.0.0
```

Pro Version (requires license token):

```
docker run -d --name localstack_test \  
-e SERVICES=s3,ses \  
-e DEFAULT_REGION=ap-northeast-1 \  
-e LOCALSTACK_AUTH_TOKEN=your-token-here \  
-p 4566:4566 \  
localstack/localstack:latest
```

Verify running:

```
docker ps
```

Step 2 - Configure AWS CLI Profile

```
aws configure --profile localstack  
AWS Access Key ID:     test  
AWS Secret Access Key: test  
Default region name:   ap-northeast-1  
Default output format: json
```

Step 3 - S3 Commands

Set checksum env vars first (run every new PowerShell window):

```
$env:AWS_REQUEST_CHECKSUM_CALCULATION="when_required"  
$env:AWS_RESPONSE_CHECKSUM_VALIDATION="when_required"
```

Create bucket:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
s3 mb my-test-bucket
```

```
s3 mb s3://my-test-bucket --region ap-northeast-1
```

List buckets:

```
aws --profile localstack --endpoint-url=http://localhost:4566 s3 ls
```

Upload file:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
s3 cp testfile.txt s3://my-test-bucket/
```

List files in bucket:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
s3 ls s3://my-test-bucket/
```

Download file:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
s3 cp s3://my-test-bucket/testfile.txt downloaded.txt
```

Delete file:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
s3 rm s3://my-test-bucket/testfile.txt
```

Step 4 - SES Commands

NOTE: Re-verify email every time LocalStack restarts!

Free Version (SESv1):

Verify sender email:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
ses verify-email-identity \  
--email-address sender@example.com --region ap-northeast-1
```

List identities:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
ses list-identities --region ap-northeast-1
```

Send email:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
ses send-email \  
--from sender@example.com \  
--destination "ToAddresses=recipient@example.com" \  
--message "Subject={Data=Test Subject,Charset=UTF-8},Body={Text={Data=Hello from  
LocalStack,Charset=UTF-8}}" \  
--region ap-northeast-1
```

Check send statistics:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
ses get-send-statistics --region ap-northeast-1
```

Pro Version (SESv2):

Verify sender email:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
sesv2 create-email-identity \  
--email-identity sender@example.com --region ap-northeast-1
```

List identities:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \  
sesv2 list-email-identities --region ap-northeast-1
```

Send email:

```
aws --profile localstack --endpoint-url=http://localhost:4566 \
  sesv2 send-email \
  --from-email-address sender@example.com \
  --destination "ToAddresses=recipient@example.com" \
  --content "Simple={Subject={Data=Test Subject, Charset=UTF-8}, Body={Text={Data=Hello, Charset=UTF-8}}}" \
  --region ap-northeast-1
```

Step 5 - Auto Init Script (init-aws.ps1)

Run this after every LocalStack start to restore S3 bucket and SES verified email:

```
Start-Sleep -Seconds 5
$env:AWS_REQUEST_CHECKSUM_CALCULATION="when_required"
$env:AWS_RESPONSE_CHECKSUM_VALIDATION="when_required"
aws --profile localstack --endpoint-url=http://localhost:4566 \
  s3 mb s3://my-test-bucket --region ap-northeast-1
aws --profile localstack --endpoint-url=http://localhost:4566 \
  ses verify-email-identity \
  --email-address sender@example.com --region ap-northeast-1
Write-Host "LocalStack initialized successfully!"
```

Step 6 - Docker Compose

```
version: '3.8'
services:
  localstack:
    image: localstack/localstack:3.0.0
    container_name: localstack_test
    ports:
      - "4566:4566"
    environment:
      - SERVICES=s3,ses
      - DEFAULT_REGION=ap-northeast-1
      - AWS_DEFAULT_REGION=ap-northeast-1
      - AWS_ACCESS_KEY_ID=test
      - AWS_SECRET_ACCESS_KEY=test
      - DEBUG=1
    volumes:
      - ./init-scripts:/etc/localstack/init/ready.d
```

Step 7 - .NET 8.0 Integration

NuGet Packages:

```
dotnet add package AWSSDK.S3
dotnet add package AWSSDK.SimpleEmail
dotnet add package AWSSDK.Extensions.NETCore.Setup
```

appsettings.Development.json:

```
{
  "AWS": {
    "ServiceURL": "http://localhost:4566",
    "Region": "ap-northeast-1",
    "AccessKey": "test",
    "SecretKey": "test",
    "ForcePathStyle": true
  },
}
```

```
"S3": { "BucketName": "my-test-bucket" },  
"SES": { "SenderEmail": "sender@example.com" }  
}
```

Important .NET settings:

- * ForcePathStyle = true (required for S3)
- * DisablePayloadSigning = true (required for S3 uploads)
- * ChecksumAlgorithm = ChecksumAlgorithm.NONE (required for S3 uploads)

Key Notes

- * Use localstack:3.0.0 for free - latest requires paid license
- * SESv1 (ses commands) for free version
- * SESv2 (sesv2 commands) for pro version
- * LocalStack does NOT send real emails - simulates API only
- * All data (S3, SES) is lost when container restarts
- * Set checksum env vars in every new PowerShell window before S3 uploads
- * No AWS account needed

Quick Start Checklist

1. Open Docker Desktop
2. Run: docker-compose up -d
3. Run: docker ps (verify healthy)
4. Run: .\init-aws.ps1 (create bucket + verify email)
5. Set checksum env vars in PowerShell
6. Start coding / testing!